Johannes Roth

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Applied AI / Machine Learning Engineer and PhD researcher with 6+ years of experience turning ambiguous data problems into scalable ML systems, evaluation workflows, and production-facing tools. Background spans large-scale computer vision infrastructure, dataset curation over billions of images, production ML services for travel products, and business information systems. Strong in Python, PyTorch, computer vision, representation learning, data quality, model evaluation, and cross-functional product engineering. Open to industry roles from **September 2026**.

EXPERIENCE

PhD Researcher - NeuroAI / Computational Cognitive Neuroscience ***May 2022 – Present***
Max Planck Institute for Human Cognitive and Brain Sciences & University of Giessen

- Researching applied NeuroAI methods for efficient experimental design: translating open-ended scientific questions into validation-set design, active sampling, model evaluation, and scalable data-selection workflows.
- Built the ML/data infrastructure behind these workflows, including public datasets, embedding-search utilities, **AWS S3/IAM**-backed data access, and re:vision challenge infrastructure for external research teams.
- Processed **2.1B** ReLAION images to score naturalness, identify approximately **500M** photographic images, and release **ReLAION-2B Natural** with a **167GB** model-feature release.
- Developed hard-example mining and evaluation workflows using model features, nearest-neighbor search, mini-batch k-Means, and Ridge encoding models to study dataset coverage and model–brain generalization.

Research Assistant - ML in Medicine ***Jun 2021 – May 2022***
ScaDS.AI Dresden/Leipzig

- Built reproducible deep learning workflows for medical image segmentation and clinical prediction, covering preprocessing, GPU training, model comparison, and uncertainty-aware evaluation.
- Engineered a multi-plane **UNet++** ensemble for glioblastoma segmentation and developed **FT-Transformer/SAINT** clinical prediction models with epistemic uncertainty estimation.

Data Scientist / ML Engineer (Working Student) ***Oct 2019 – May 2021***
CHECK24 (Travel Vertical)

- Built applied ML and backend features for a consumer travel product, including a Flask/Redis service for fast image inference across **>20 million** hotel images.
- Used model outputs for deduplication, retrieval, classification, quality scoring, and recommendation tuning; optimized serving with **ONNX** and worked with **GCP BigQuery** and **Grafana** for product-data workflows.
- Worked cross-functionally on booking-engine features, hotel-price outlier detection, ranking evaluation, Bayesian search, and image-quality pipelines used by product and data teams.

Data Scientist (Working Student) ***Oct 2018 – Oct 2019***
Webdata Solutions GmbH (now Vistex)

- Worked on product matching for e-commerce data, using **Azure on-demand servers** while turning noisy scraped web data into usable training and evaluation datasets.
- Rebuilt parts of the matching pipeline with a neural-network approach, improving matching accuracy from **<50%** to **92%**.

Full-stack Developer (Freelance) ***Oct 2020 – May 2021***
Kimetric UG

- Delivered full-stack web projects for academic clients, from requirements clarification and backend implementation to deployment.
- Built Django-based websites, configured Linux servers with Nginx and Gunicorn, and wrote deployment scripts for reliable handoff.

PROJECTS & OPEN SOURCE

- **thingsvision (Core Contributor)** – Unified Python API for extracting and comparing image representations from **100+** vision and foundation models; **460k+** PyPI downloads.
- **ReLAION-2B Natural** – Naturalness scores, approximately **500M** photographic images, and a **167GB** feature release for data selection, retrieval, and model-comparison workflows.
- **LAION-fMRI / re:vision** – Stimulus selection and release infrastructure for a deeply sampled **7T fMRI** dataset with **25,052** natural images and **165** scan sessions.

EDUCATION

PhD Candidate - Computer Science / Machine Learning ***May 2022 – Present***
Max Planck Institute for Human Cognitive and Brain Sciences & University of Giessen
Dissertation on efficient experimental design, large-scale visual datasets, and model–brain alignment for computational neuroscience.

M.Sc. Computer Science ***2017 – 2021***
Leipzig University
Grade **1.2**. Focus: Data Analysis, Machine Learning, Medical Image Processing.

- **Master's Thesis (Grade 1.1)** – Used GANs and neural encoding models to synthesize stimuli that maximally activate targeted brain regions, recovering known category-selective areas in human visual cortex.

B.Sc. Business Information Systems ***2014 – 2017***
Leipzig University
Grade **1.5**. Focus: Distributed Systems, E-Commerce, Data Management, Economics.

TECHNICAL SKILLS

Applied AI & Vision	Computer vision, representation learning, model–brain comparison, fMRI workflows
ML & Data Systems	Dataset curation, billion-scale filtering, embedding search, data quality, active sampling, validation data design, hard-example mining
Training & Evaluation	PyTorch, TensorFlow, scikit-learn, Transformers, automated model comparison, ranking metrics, uncertainty estimation, reproducible experiment workflows
Models	CLIP/OpenCLIP-style embeddings, ViT, CNNs, UNet++, GANs, foundation-model feature extraction and representation comparison
Engineering	Python, SQL, Docker, ONNX, AWS S3/IAM, GCP BigQuery, Azure, Git, CI/CD, Flask, Redis, Linux servers, Nginx/Gunicorn, Bash, SLURM/HPC, Grafana
Communication	Technical writing, research talks, stakeholder alignment, mentoring, German (Native), English (Fluent)

SELECTED PUBLICATIONS & AWARDS

- **Award (2025):** CMBB Replication Award for reliable coding practices in neuroscience.
- **F.P. Mahner*, J. Roth***, et al. *Characterizing Universal Object Representations Across Vision Models*. arXiv preprint, 2026 (shared first author, under review).
- **J. Roth**, M. N. Hebart. *How to sample the world for understanding the visual system*. CCN 2025 Oral.
- **J. Roth** et al. *Ten principles for reliable, efficient, and adaptable coding*. Communications Psychology 2025; **22k+** accesses.

LEADERSHIP & COMMUNITY

- **PhD Representative (2023–2024):** Elected to represent **>180** doctoral researchers at MPI CBS.
- **Mentoring:** Supervised working students and interns in the research group; translated ambiguous research goals into scoped engineering tasks and feedback loops.
- **Research communication:** Presented work at international conferences, institute colloquia, interdisciplinary research meetings, and public ML outreach events.